Electric car: launch the design

antocciphysics

UNIT 8 / DAY 1 / CPA

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Plan a safe team workflow and identify the car's energy pathway.

SUCCESS CRITERIA - I can...

1. Label the battery, switch, motor and connections.
2. Identify safety checkpoints before tools or power.
3. Connect a design choice with a measurable outcome.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Warm-up: bring back an earlier idea

A stationary coil has constant magnetic flux 0.003 Wb through each turn. Does that constant flux produce sustained induced emf? Explain.

Marzano check - circle: ① ② ③

02 Worked example: explain the model

Reference: average speed = distance/elapsed time. Use your own design and measurements. Battery → motor → drivetrain. Follow teacher training for tools and hot materials.

Worked reasoning: A design claim becomes testable when it names a measurement. "The car is fast" can become "The car travels 3 m in under 2 s on this level track." That condition corresponds to average speed above 1.5 m/s.

Your turn: Draw a test track with start, finish, distance and timing method labeled.

Start Finish



Distance: ____ Timing method: _____

Marzano check - circle: ① ② ③

03 Read and interpret the model

Sketch your proposed car. Label the battery, switch, motor, driven axle and wheels. Add a theme that will not obstruct moving parts.



Top view: the large rectangle is the chassis; small rectangles are wheels; the two crossbars are axles.

Marzano check - circle: ① ② ③

04 Apply the physics

Set up your build team: choose a build lead, safety checker and recorder. Complete the role table. Identify the teacher checkpoints required before cutting, drilling, soldering and the first powered test. Leave approvals blank until they happen.

Role	Team member	Approval / checkpoint
Build lead	____	____
Safety checker	____	____
Recorder	____	____

Marzano check - circle: ① ② ③

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Continue down each column.

05 Explain your reasoning

Write one measurable design goal and one constraint. Explain how you will measure the goal instead of judging only by appearance.

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

An ideal series loop has supply 24 V. One component drops 4 V. Find the other component's drop.
Draw the stated circuit and label quantities relevant to the question.

Draw the complete circuit and label shared nodes.

Marzano check - circle: 1 2 3

07 Exit check

Complete: The part that converts electrical energy into rotation is ____. Before a powered test, I will check ____.

Check / criterion	Measurement	Decision rule
___	___	___
___	___	___

Marzano check - circle: 1 2 3

08 STRETCH: test the model

Propose a second success criterion that could conflict with speed, such as straight tracking or reliability. Describe how both would be measured.

Check / criterion	Measurement	Decision rule
___	___	___
___	___	___

Marzano check - circle: 1 2 3